UNIVERSITÀ DEGLI STUDI DI UDINE

hic sunt futura

# Reticular Local Abstraction (RLA) Conceptual Model, Epistemic CNNs and a PCE bridge

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Università degli Studi di Udine A.A.  2024-2025

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# Introduction

# 1. INTRODUCTION

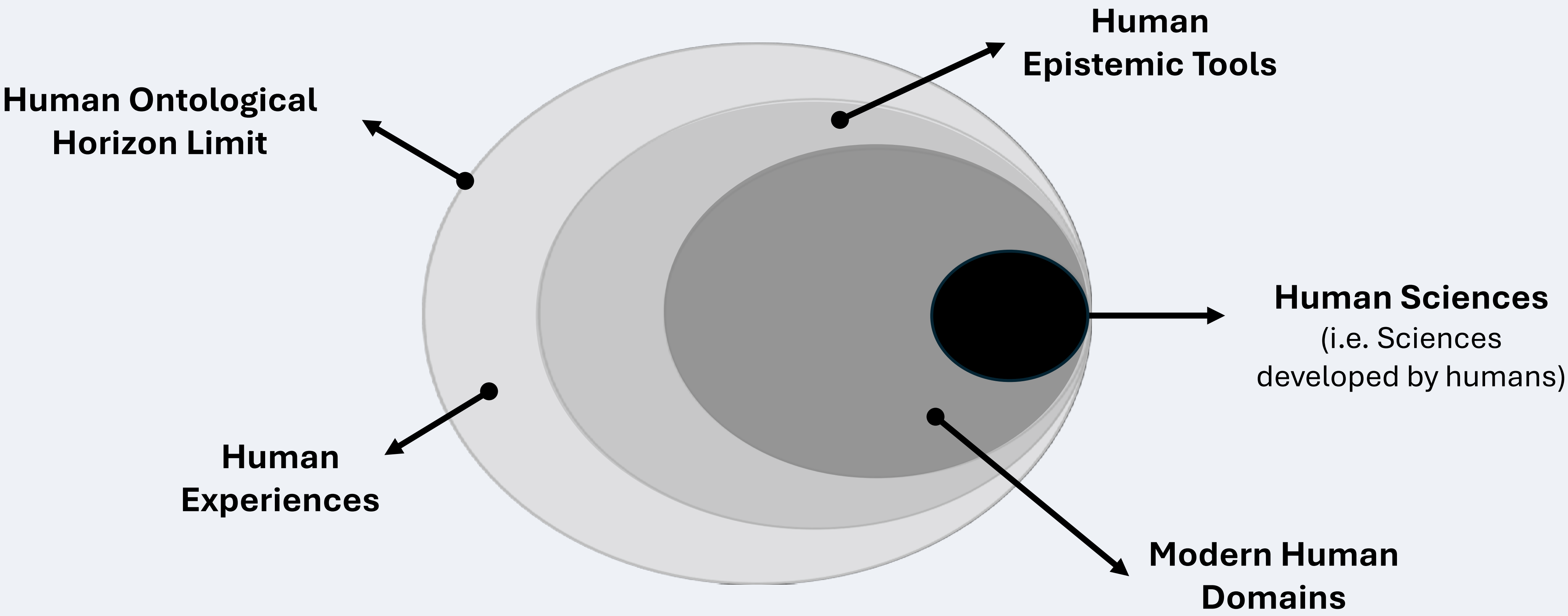
## DOMAINS OF HUMAN EXPERIENCE

| Domain | Core Idea | Key References |
|---|---|---|
| **Physical–Natural** | Objective order, causality, empirical lawfulness | Kant 1781; Popper 1959; Kuhn 1962; Lakatos 1978 |
| **Technical–Technological** | Reason applied to creation of artifacts and systems | Descartes 1637; Simon 1969; Floridi 2010; Boon 2018 |
| **Social–Human** | Rational action, normativity, historical mediation | Hegel 1807; Habermas 1981; Bhaskar 1979; Weimer 2022 |
| **Subjective–Phenomenological** | Inner experience, intentionality, self-awareness | Descartes 1641; Husserl 1913; Merleau-Ponty 1945; Nagel 1974 |
| **Intersubjective–Cultural** | Shared meaning, language, symbolic worlds | Wittgenstein 1953; Latour 2005; Barad 2007 |

A **domain** is a distinct mode of human experience and action — defined by what can be *known*, *done*, and *meant* within its internal logic and limits.

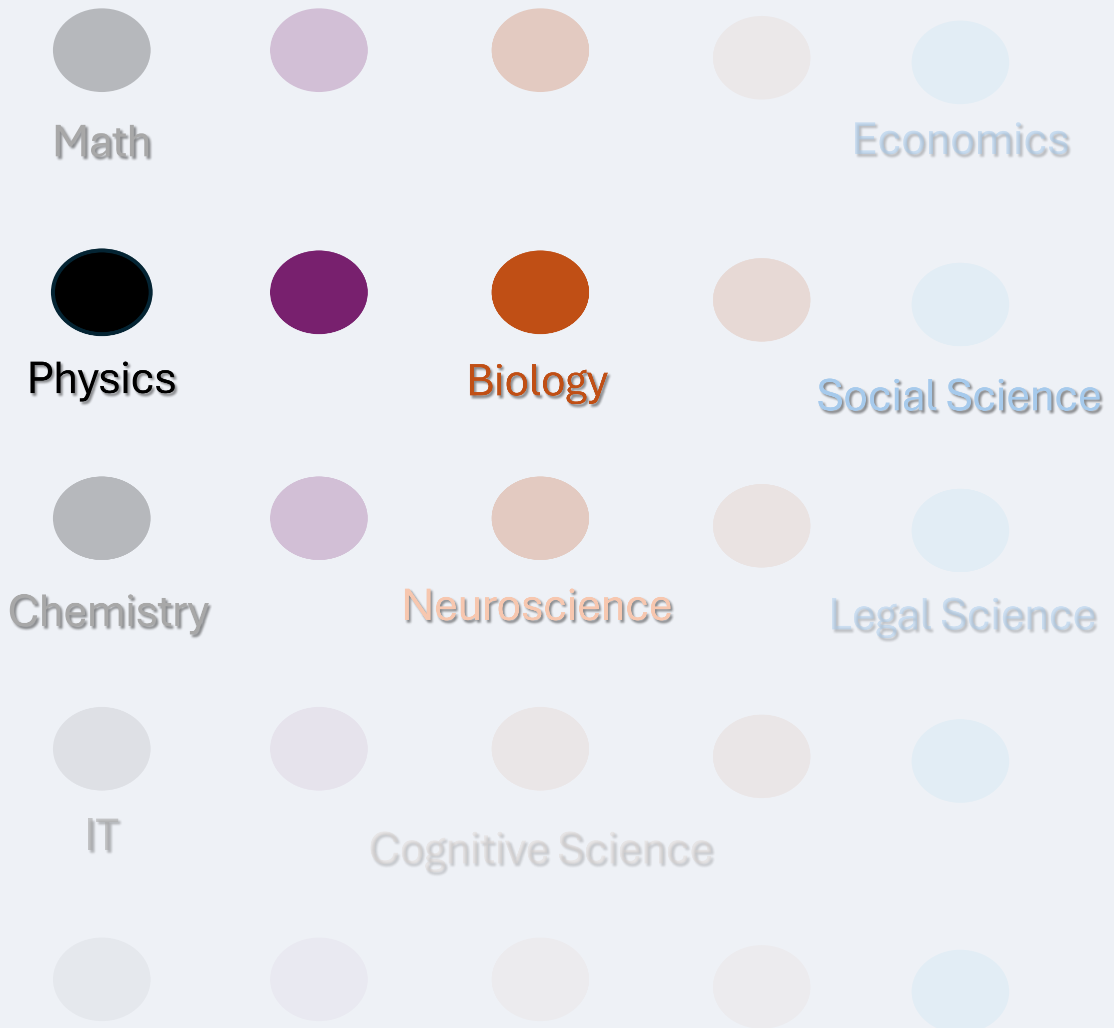
# 1. INTRODUCTION

## DOMAINS OF HUMAN EXPERIENCE



# 1. INTRODUCTION

## HUMAN SCIENCES



# 1. INTRODUCTION

## HUMAN SCIENCES

Physics

Chemistry

Biology

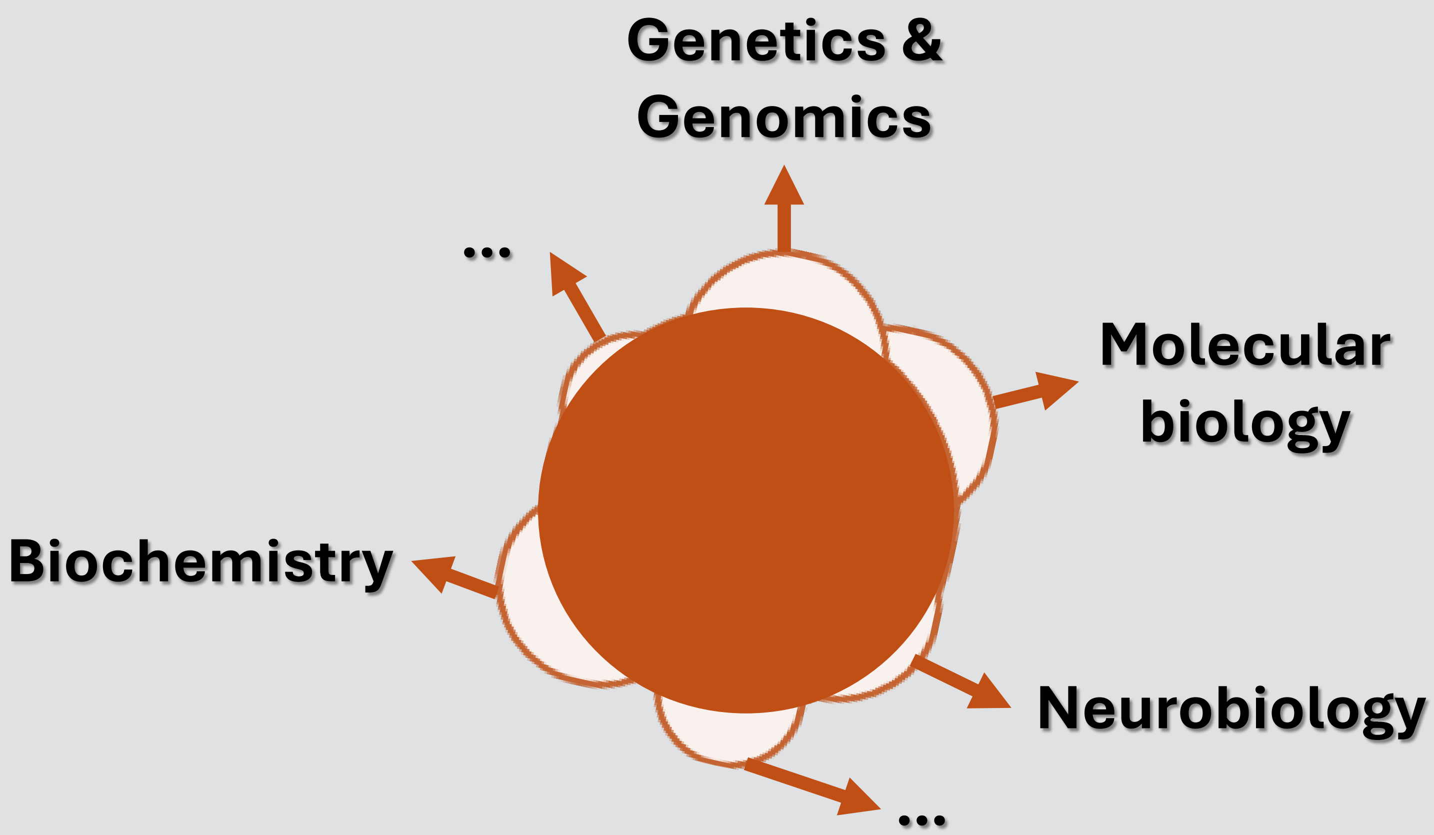
# 1. INTRODUCTION

## HUMAN SCIENCES

- Physics
- Chemistry
- Biology

Chemistry

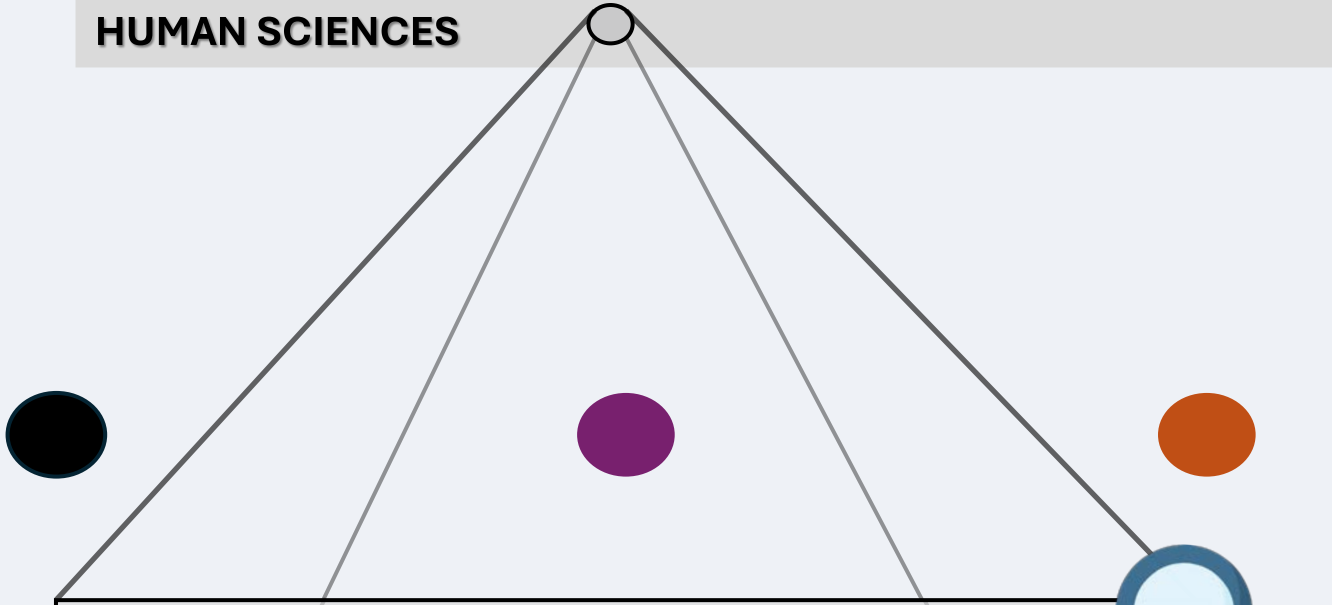
Biology


Genetics &
Genomics
...
Molecular
biology
Biochemistry
Neurobiology
...

# Biology

# 1. INTRODUCTION

## HUMAN SCIENCES



- ***Are the sciences orderable?***
- ***Does it make sense to conceive an ordering of scientific disciplines.***
- ***If so, is such an order absolute or only relative (local, contextual)?***

# 1. INTRODUCTION

## HUMAN SCIENCES

| View | Epistemic Structure | Key Authors (Date) | Type of Order |
| --- | --- | --- | --- |
| **Classical** | Linear, foundational | Kant (1781), Comte (1830–42) | **Absolute** |
| **Modern** | Contextual, incommensurable | Kuhn (1962), Feyerabend (1975), Lakatos (1978) | **Relative / Local** |
| **Contemporary** | Relational, dynamic | Latour (2005), Barad (2007), Floridi (2010) | **Topological / Emergent** |

**The sciences form not a hierarchy, but a reticulum — an evolving topology of domains linked by partial translatability.**

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# RLA Concept Model

# 2. RLA Concept Model

## CONTIGUOUS DOMAINS

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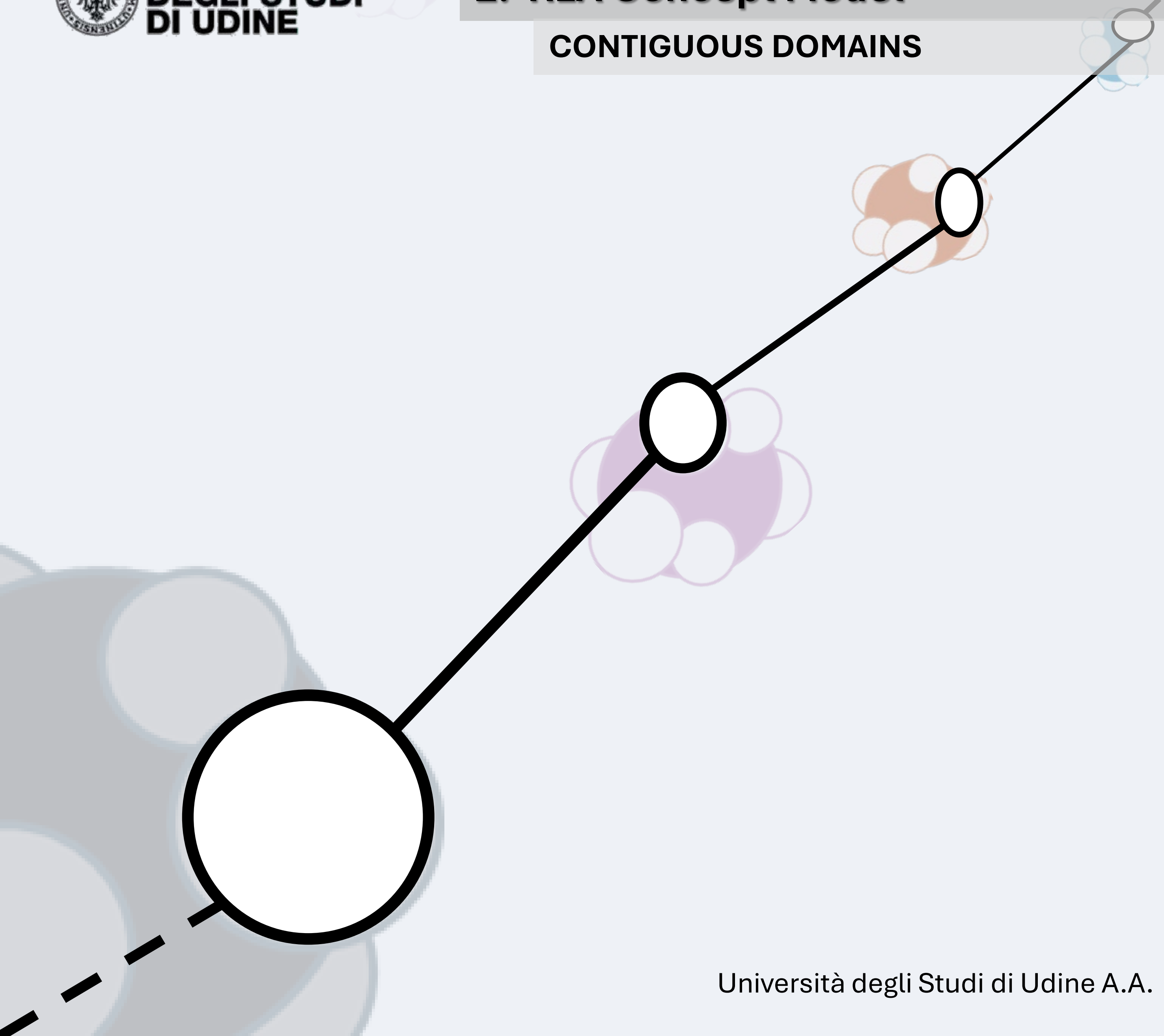
## 2. RLA Concept Model

### CONTIGUOUS DOMAINS



## 2. RLA Concept Model

### CONTIGUOUS DOMAINS

**Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. Chicago: University Chicago Press.**
*Paradigms as local frameworks; partial translatability between disciplines defines epistemic contiguity.*

**Cartwright, N. (1983). *How the Laws of Physics Lie*. Oxford: Clarendon Press.**
*Scientific laws are locally valid; bridge laws connect overlapping domains.*

**Noble, D. (2012). *A theory of biological relativity: No privileged level of causation. Interface Focus, 2*(1), 55–64.**
*No causal hierarchy; biological levels are reciprocally connected — model of epistemic contiguity.*

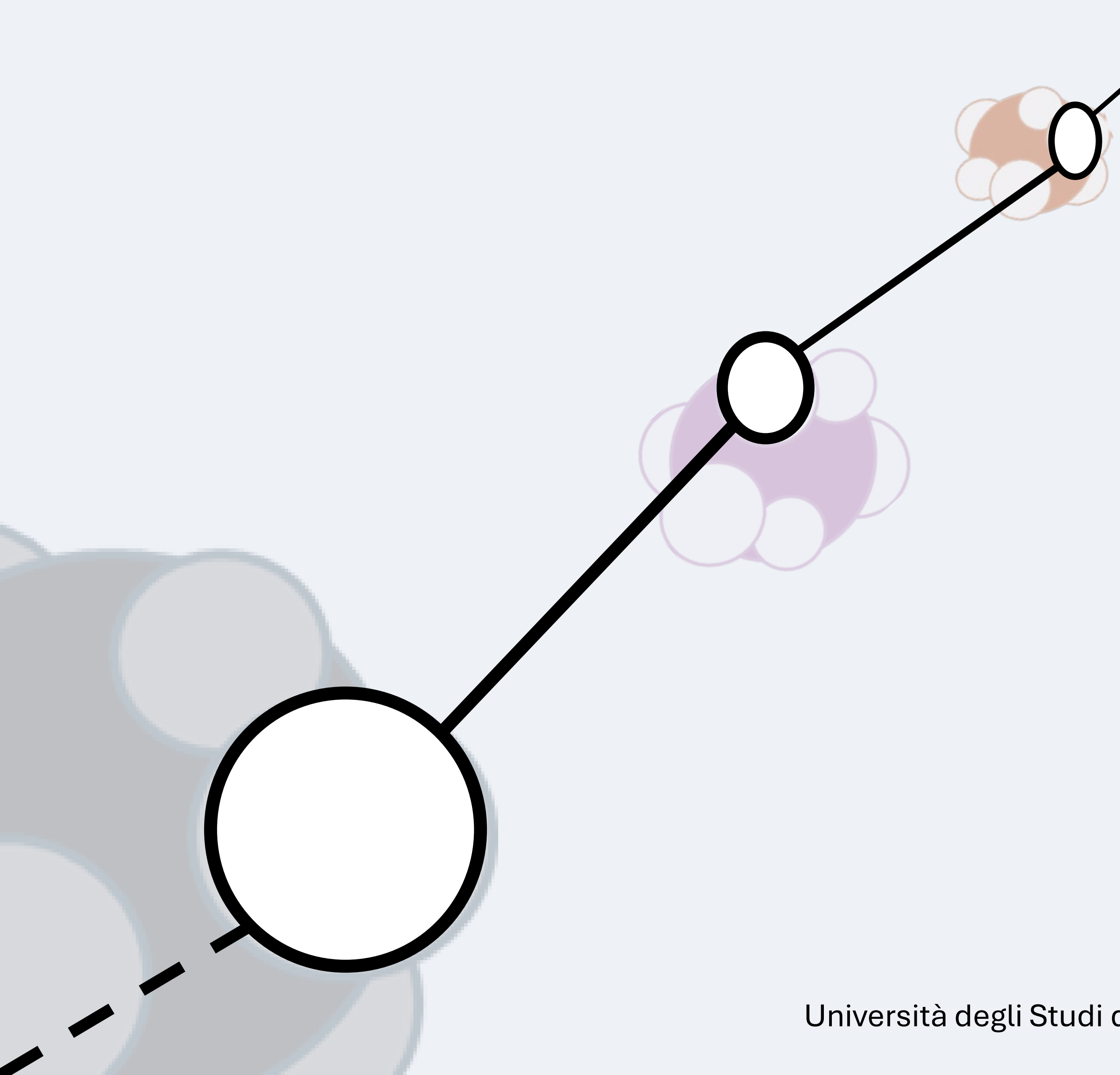
**Humphreys, P. (2016). *Emergence: A Philosophical Account*. Oxford: Oxford University Press.**
*Inter-level dependence formalizes domain adjacency and controlled emergence.*

**Bechtel, W., & Abrahamsen, A. (2005). *Explanation: A mechanist alternative. Studies in History and Philosophy of Biological and Biomedical Sciences, 36*(2), 421–441.**
Mechanistic cross-level integration as empirical bridge between models.

**Kauffman, S. A. (2000). *Investigations*. Oxford: Oxford University Press.**
Reticulate causation and “adjacent possible” describe emergence through non-injective transitions.

# 2. RLA Concept Model

## CONTIGUOUS DOMAINS



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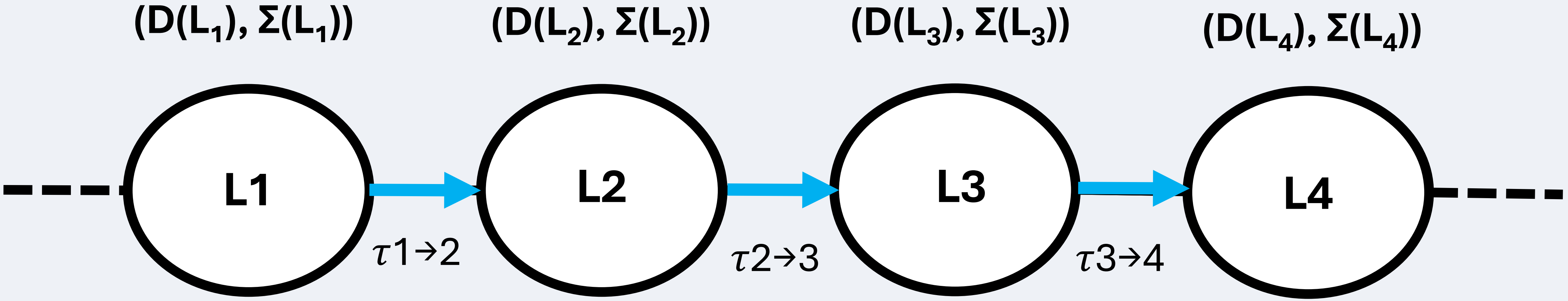
## 2. RLA Concept Model

### CONTIGUOUS DOMAINS




$(D(L_n), \Sigma(L_n))$

$\tau\, n \to n+1$

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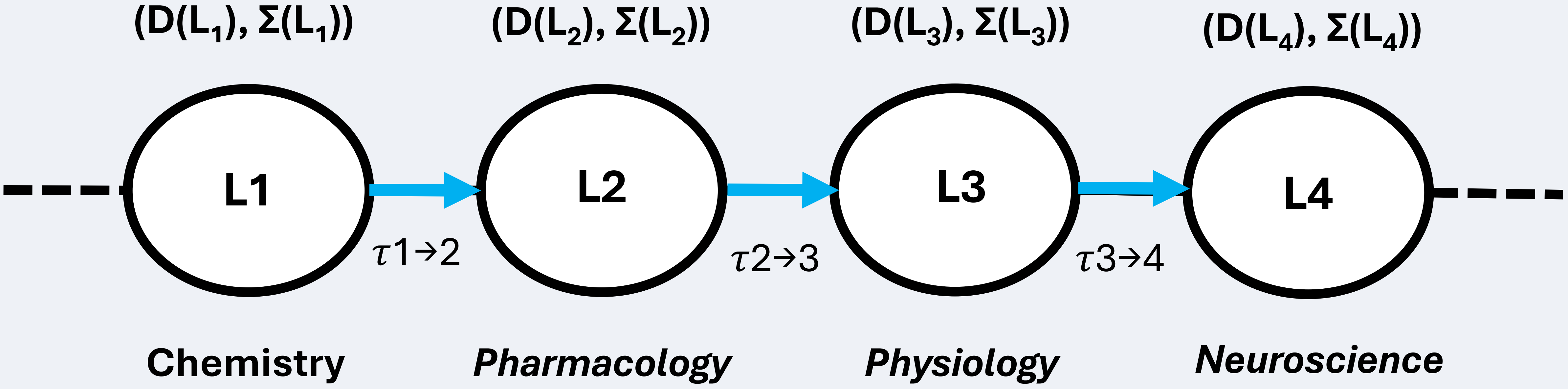
## 2. RLA Concept Model

### CONTIGUOUS DOMAINS

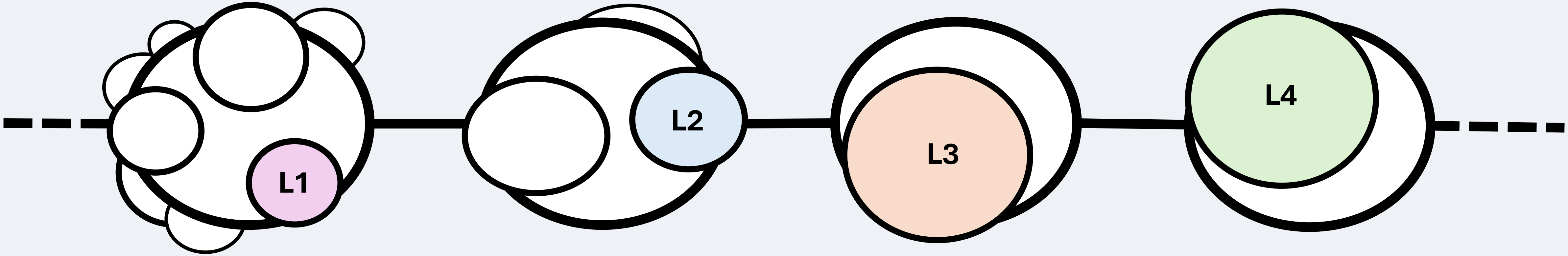



**Conventional Linearization Only**

## 2. RLA Concept Model

### CONTIGUOUS DOMAINS

| Chemistry | *Pharmacology* | *Physiology* | *Neuroscience* |
|---|---|---|---|




| Organic / Molecular Chemistry | Molecular Pharmacology | Systemic Physiology | Neuroethology / Psychobiology |
|---|---|---|---|

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# 2. RLA Concept Model

## CONTIGUOUS DOMAINS

L1 — L2 — L3 — L4

### $L_1$ Organic/Molecular Chemistry

- Domain: Organic & quantum chemistry; structure–function of small molecules.
- Core laws: Quantum mechanics (Schrödinger, DFT), thermodynamics (Gibbs, Arrhenius), chemical kinetics.

***Dominant References:*** *Atkins & de Paula, Physical Chemistry (OUP, 2023) Leach, Molecular Modelling (Pearson, 2013) Cramer, Essentials of Computational Chemistry (Wiley, 2013)*

### $L_2$ Molecular Pharmacology

- Domain: Drug–target interactions, receptor binding, enzyme inhibition, signaling.
- Core laws: Langmuir binding, Hill equation, Michaelis–Menten kinetics, receptor theory.

***Dominant References:*** *Rang & Dale's Pharmacology (Elsevier, 2024) Kenakin, Pharmacology in Drug Discovery (Academic Press, 2016) Gabrielsson & Weiner, Pharmacokinetic and Pharmacodynamic Data Analysis (CRC, 2012)*

### $L_3$ Systemic Physiology

- Domain: Integrated organ function, homeostasis, metabolic and hormonal control.
- Core laws: Mass balance ODEs, feedback regulation, compartment and control models.

***Dominant References:*** *Guyton & Hall, Textbook of Medical Physiology (Elsevier, 2021) Noble, The Music of Life (OUP, 2006) Kitano, Systems Biology (Science, 2002)*

### $L_4$ Neuroethology

- Domain: Neural circuits underlying perception, motivation, and adaptive behavior.
- Core laws: Neural population dynamics (Hopfield, Wilson–Cowan), reinforcement learning, drift–diffusion decision models.

***Dominant References:*** *Kandel et al., Principles of Neural Science (McGraw-Hill, 2021)Schultz et al., Science 275 (1997) — dopamine reward prediction Tinbergen, The Study Instinct (OUP,1951)*

| Transition | Type |
|---|---|
| $\tau 1 \rightarrow 2$ | [...] |

Formalism
[...]

| Transition | Type |
|---|---|
| $\tau 1 \rightarrow 2$ | [...] |

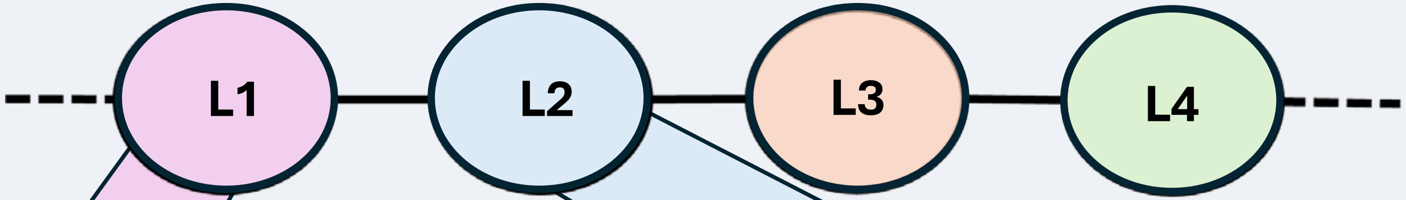
Formalism
[...]

| Transition | Type |
|---|---|
| $\tau 1 \rightarrow 2$ | [...] |

Formalism
[...]

# 2. RLA Concept Model

## CONTIGUOUS DOMAINS




### $L_1$ Organic/Molecular Chemistry

**Domain**: Small molecules; functional groups; stereochemistry; 3D conformations; charge distribution; electronic states; physicochemical descriptors (MW, logP, pKa, TPSA, HBD/HBA).

**Core Law**: Quantum/thermos + cheminformatics:

- Approx. Schrödinger/DFT; Hartree–Fock; semi-empirical MO.
- Reaction kinetics & thermodynamics: Eyring/Arrhenius; Gibbs free energy.
- Conformer generation; tautomer/protomer equilibria.
- Descriptor pipelines & QSAR/QSPR; graph-based featurization.

**Transition**

$\tau 1 \rightarrow 2$

**Type**

- Non-Injective
- Occasionally Quasi-injective

**Formalism**

Docking/MD/ΔΔG; QSAR; free-energy cycles; protonation & permeability models for ADME priors.

### $L_2$ Molecular Pharmacology

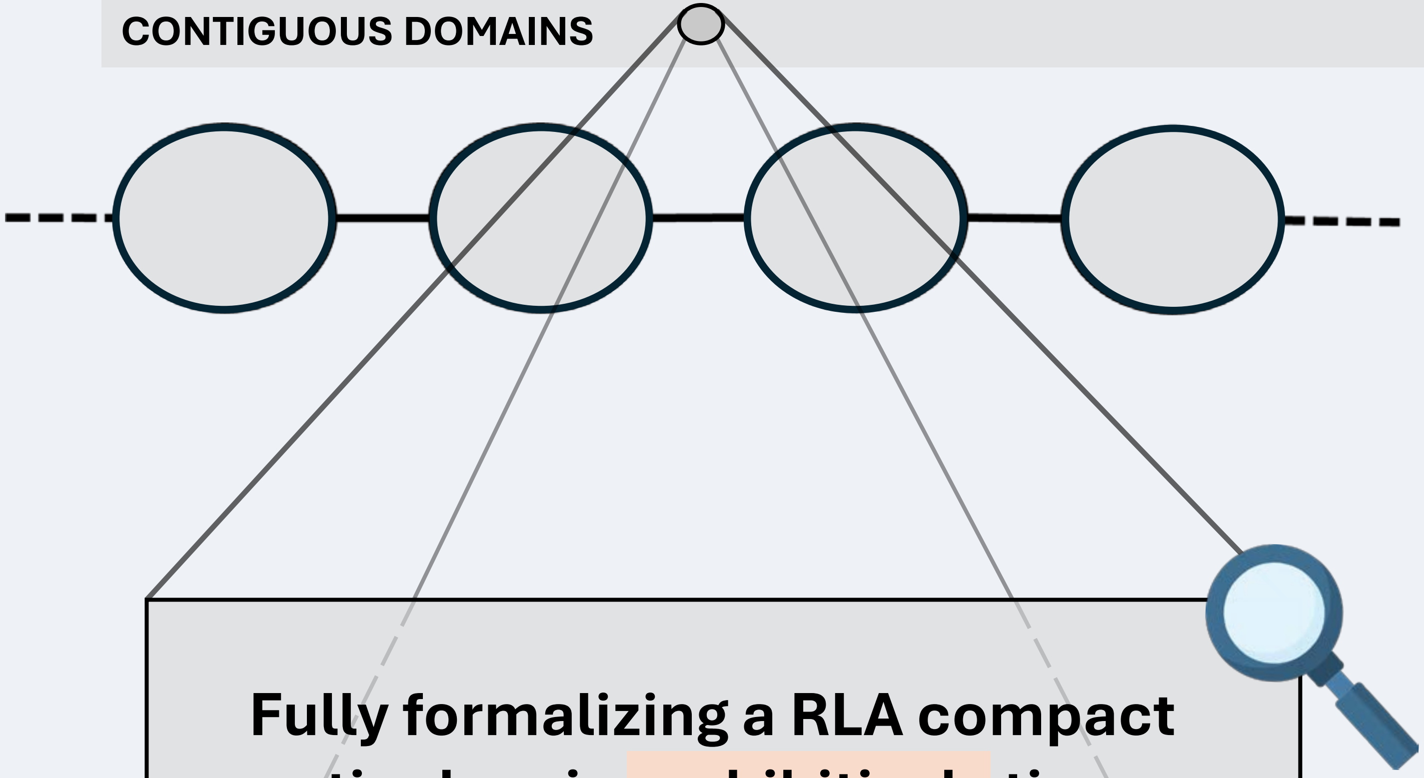
**Domain**: *Targets(receptors/ enzymes/ transporters), ligands, binding sites; affinity/efficacy; occupancy; signaling states; ADME parameters; PK/PD state.*

**Core Lawes:**

- *Receptor–ligand binding: mass-action, Langmuir, Hill; $\Delta G \leftrightarrow Kd$ΔG↔Kd.*
- *Enzyme kinetics: Michaelis–Menten, competitive/non-competitive inhibition.*
- *Allostery (Monod–Wyman–Changeux); biased agonism; signal transduction ODEs.PK models (1–3 compartment, clearance/bioavailability) and PD link (Emax, sigmoid Emax, indirect response).*

## 2. RLA Concept Model

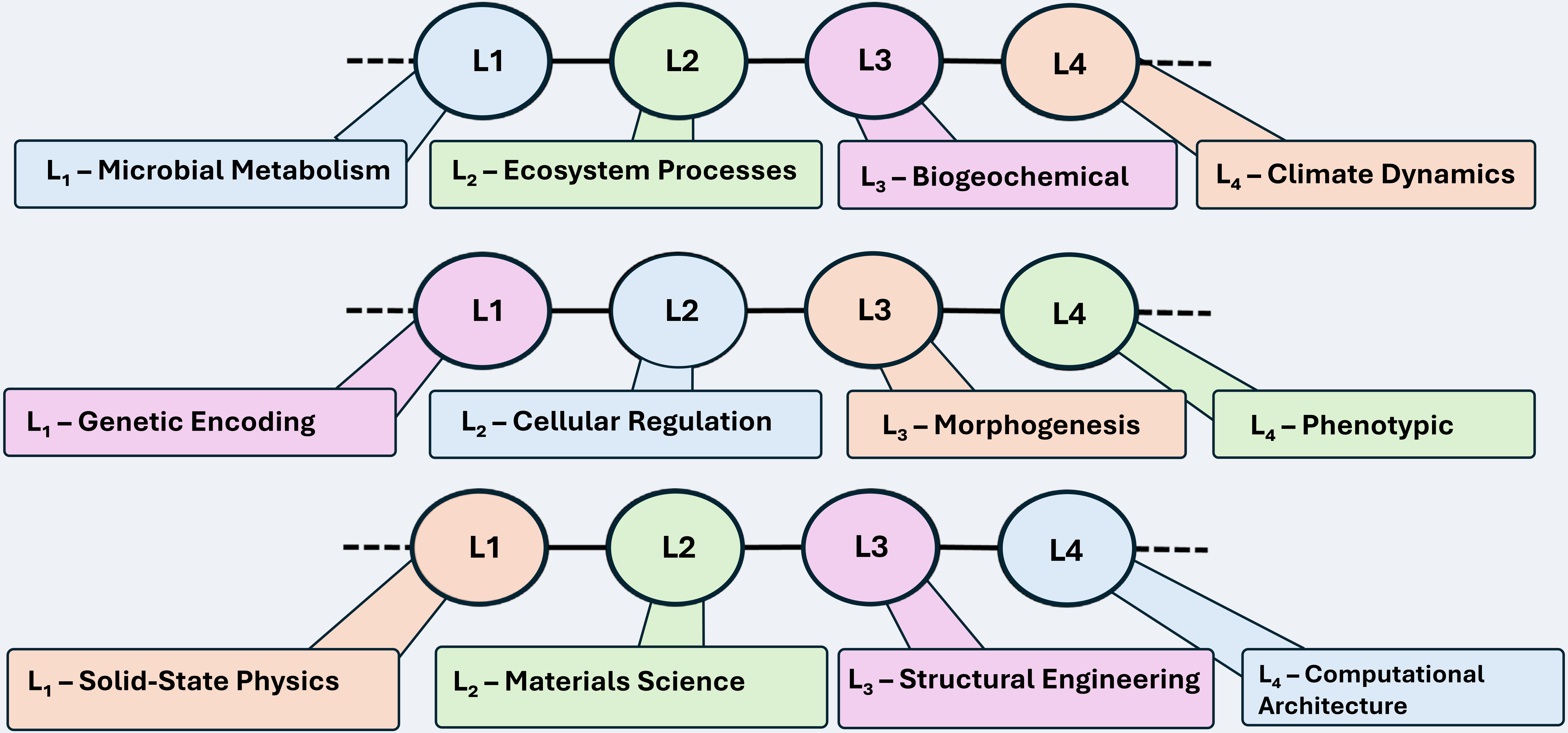
### CONTIGUOUS DOMAINS



**Fully formalizing a RLA compact reticulum is prohibitively time-consuming manually and must therefore be executed by AI under human oversight.**

## 2. RLA Concept Model

### CONTIGUOUS DOMAINS (EXAMPLES)



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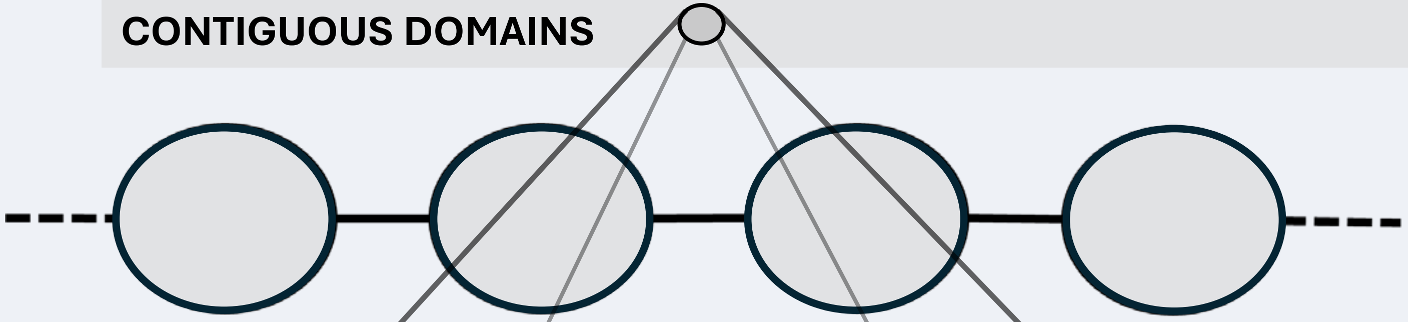
## 2. RLA Concept Model

### CONTIGUOUS DOMAINS

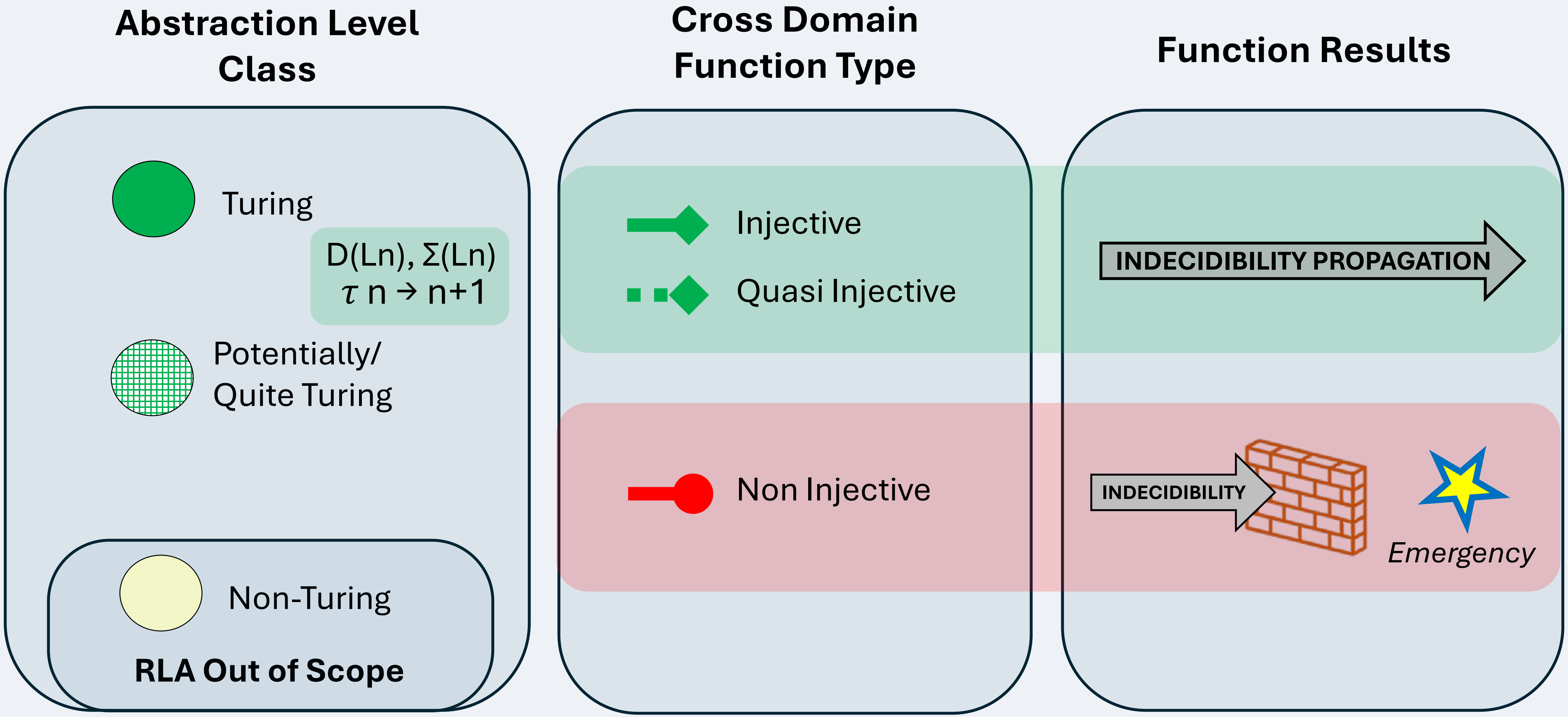


- Not all reticula are meaningful to construct;
- beyond certain domains, their formation risks becoming speculative and devoid of real significance.

It is strictly necessary to use assumptions, definitions, etc... to limit speculative risk

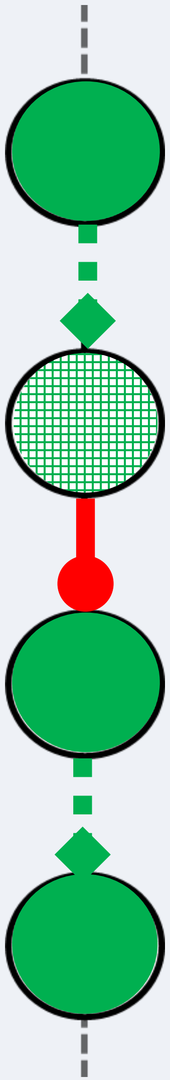
## 2. RLA Concept Model

### DEFINITIONS & ASSUMPIONS (informal, overview)



## 2. RLA Concept Model

### Biological Being RLA Representation (conceptual example)



L1 **Sistemi Fisici Discreti (Automi Cellulari)**

L2 **Microstati Molecolari (Spin-lattice)**

L3 **Chimica Computazionale (Reazioni)**

L4 **Biologia Molecolare Computazionale (DNA)**

**Scientific «Supply Chain»**

*Wolfram 2002*

*Toffoli & Margolus 1987*

*Norris 1998*

*Magnasco 1997*

*Winfree 1998*

*Benenson 2001*

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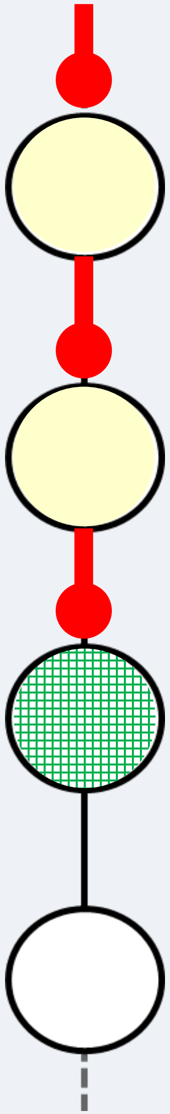
## 2. RLA Concept Model

### Biological Being RLA Representation (conceptual example)



- L5 — Cellula (Pathway Integrati)
- L6 — Tessuti e Organi Virtuali
- L7 — Reti Neurali Biologiche
- L8 — ...

**Scientific «Supply Chain»**

*Cardelli 2005*

*Noble 2008*

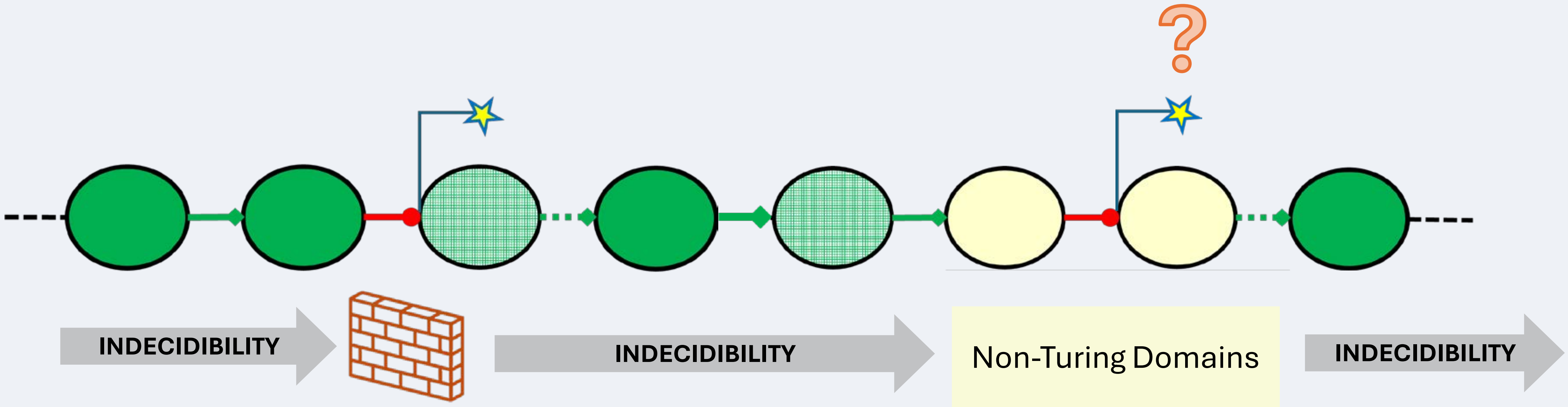
*Siegelmann & Sontag 1991*

...

**Interesting network but not "expressive" enough at every level, so it's just a conceptual exercise**

# 2. RLA Concept Model

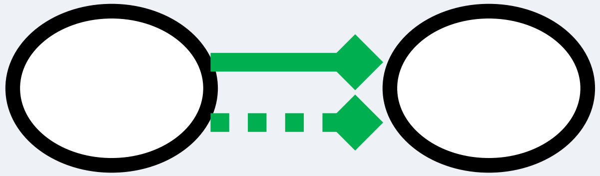
## DEFINITIONS & ASSUMPIONS (brief overview)

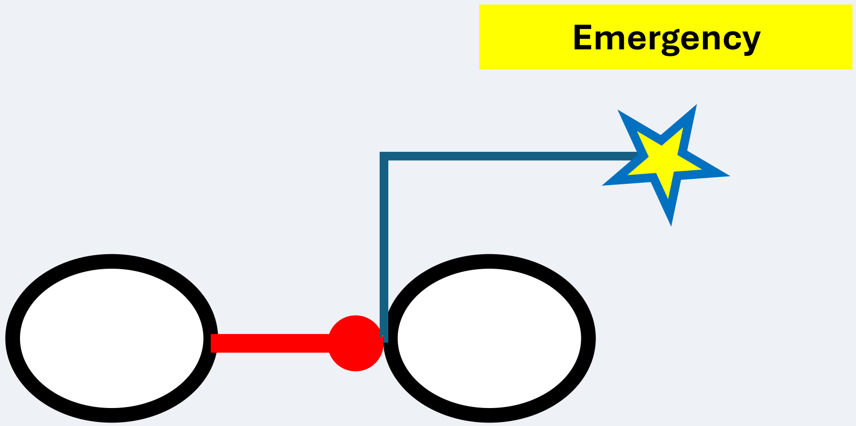



**Formalism is needed**

# 2. RLA Concept Model

## DEFINITIONS (informal, overview)





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# 2. RLA Concept Model

## FORMAL FRAMEWORK OVERVIEW

| Category | Code Range | Count | Conceptual Role |
|---|---|---|---|
| **Assumptions (ASS)** | ASS-01 → ASS-04 | **4** | Foundational premises ensuring empirical contiguity, Turing-likeness, and domain relevance of collapses |
| **Definitions (DEF)** | DEF-01 → DEF-14 | **14** | Formal constructs defining levels, transmissions, computability classes, compactness, and emergence metrics |
| **Axioms (AX)** | AX-01 → AX-04 | **4** | Primary laws governing undecidability, propagation, non-injective emergence, and epistemic closure |
| **Lemmas (LEM)** | LEM-01 → LEM-03 | **3** | Auxiliary results on composition, reactivation, and quotient interpretation of state pooling |
| **Theorems (THEO)** | THEO-01 → THEO-04 | **4** | Demonstrated consequences on propagation barriers, definability, irreversibility, and upper-level undecidability |
| **Corollaries (COR)** | COR-01 → COR-02 | **2** | Derived implications on multi-level undecidability propagation and cumulative emergence |

## 2. RLA Concept Model

### DEFINITIONS (informal, overview)

**DEF-01 — Level of abstraction.** $L = \langle D, (L)\Sigma(L)\rangle$ with computable rules acting on a state space.

**DEF-02 — Contiguity & transmission.** $\tau_{i\to i+1}: D(L_i) \to \mathcal{P}\big(D(L_{i+1})\big)$ may be non-injective and is discipline-recognized.

**DEF-03 — Turing-likeness & $C_L$ .**A level is Turing-like if it can simulate a UTM; $C_L$marks the Turing-critical subset.

**DEF-04 — (Non)injectivity classes.** Injective / quasi-injective on a subset / non-injective (collapse) are distinguished.

[…]

**DEF-09 — Observation predicates & definability.** A macro-property at $L_{i+1}$ is definable from $L_i$ if there exists a predicate on $L_i$matching all preimages; else it is emergent.

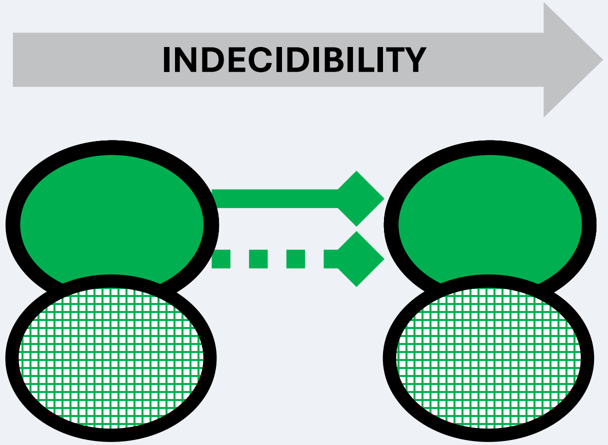
**DEF-10 — Relevance function.** A domain-specific score $s$ and threshold $\sigma$ determine when a collapse is “relevant.”

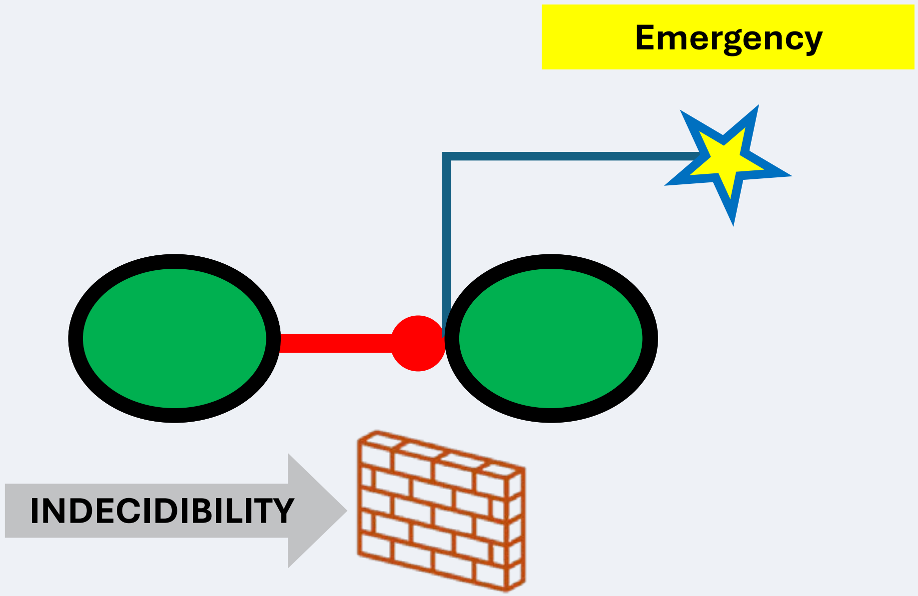
[…]

**DEF-14 — Reticular path, loop, morphism.** Paths/loops across levels; morphisms preserve levels and transmissions up to relabelling.

# 2. RLA Concept Model

## DEFINITIONS (informal, overview)



# 2. RLA Concept Model

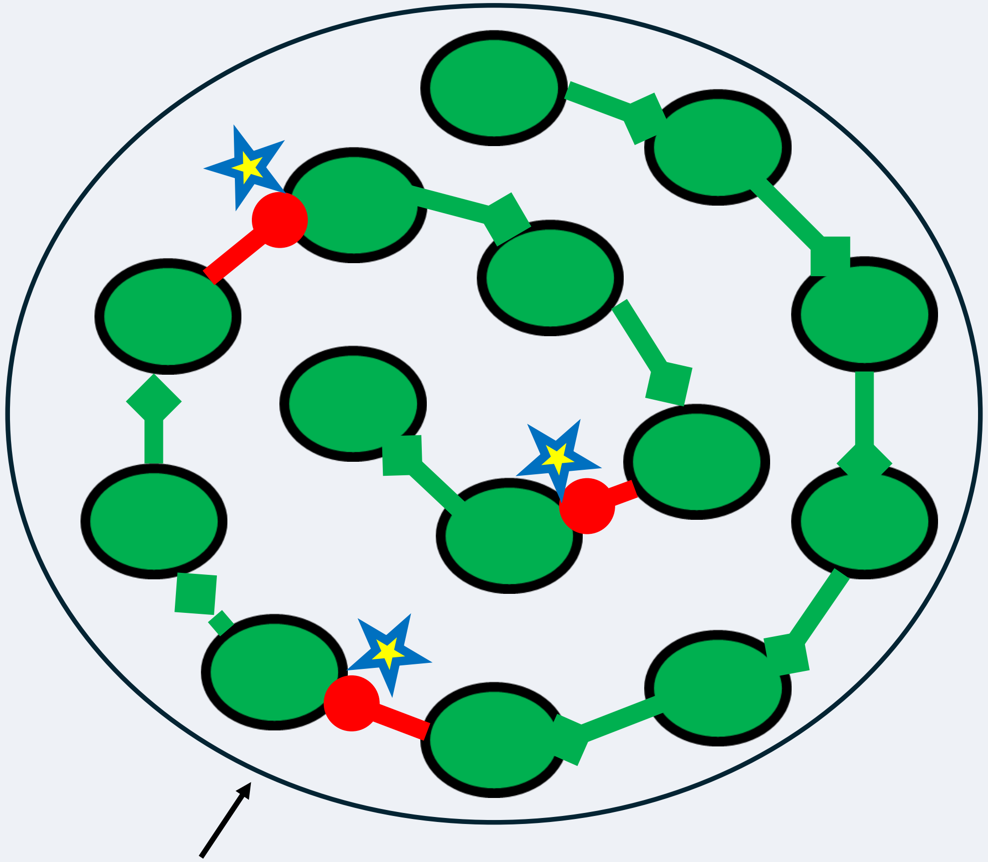
## Compat RLA Engineering

| Phase | Action | Output |
|---|---|---|
| **Scouting** | *Systematic literature review & source curation* | *Reviewed bibliographic corpus* |
| **Extraction** | *Parse rules, variables, causal links* | *Structured knowledge base* |
| **Abstraction Build** | *Generate layered* $\langle L_1 \ldots L_n \rangle$ *+* $\tau$*-mappings* | *Draft reticular hierarchy* |
| **E2E Computability** | *Every layers and every* $\tau$ *must be turing* | *Verified compact topology* |
| **Metrics** | *Compute CC, EI, RIR* | *Quantified model metrics* |

A **bibliographically anchored**, **AI-constructed**, and **epistemically validated** compact RLA topology—ready for empirical or computational deployment.

# 2. RLA Concept Model

## Compat RLA Engineering



External Collapsed Representation Horizon

- If a domain is **Turing-compatible** (DEF-07) and adjacent domains are linked by computable, empirically grounded **transmissions** (DEF-02, ASS-02), the chain forms a **compact reticulum globally computable** (DEF-05), (DEF-06), (DEF-07/08), with a **non-injective boundary** filtering external inputs (AX-04) (no external oracle, no epistemic perturbation).

- Synthetic observations within the reticular horizon are expected to be **epistemically indistinguishable** from expert human observations — a direct consequence of **undecidability propagation** (AX-01/02) and **emergence from collapse** (AX-03, THEO-02).

# 2. RLA Concept Model

## Compat RLA Engineering

[...]

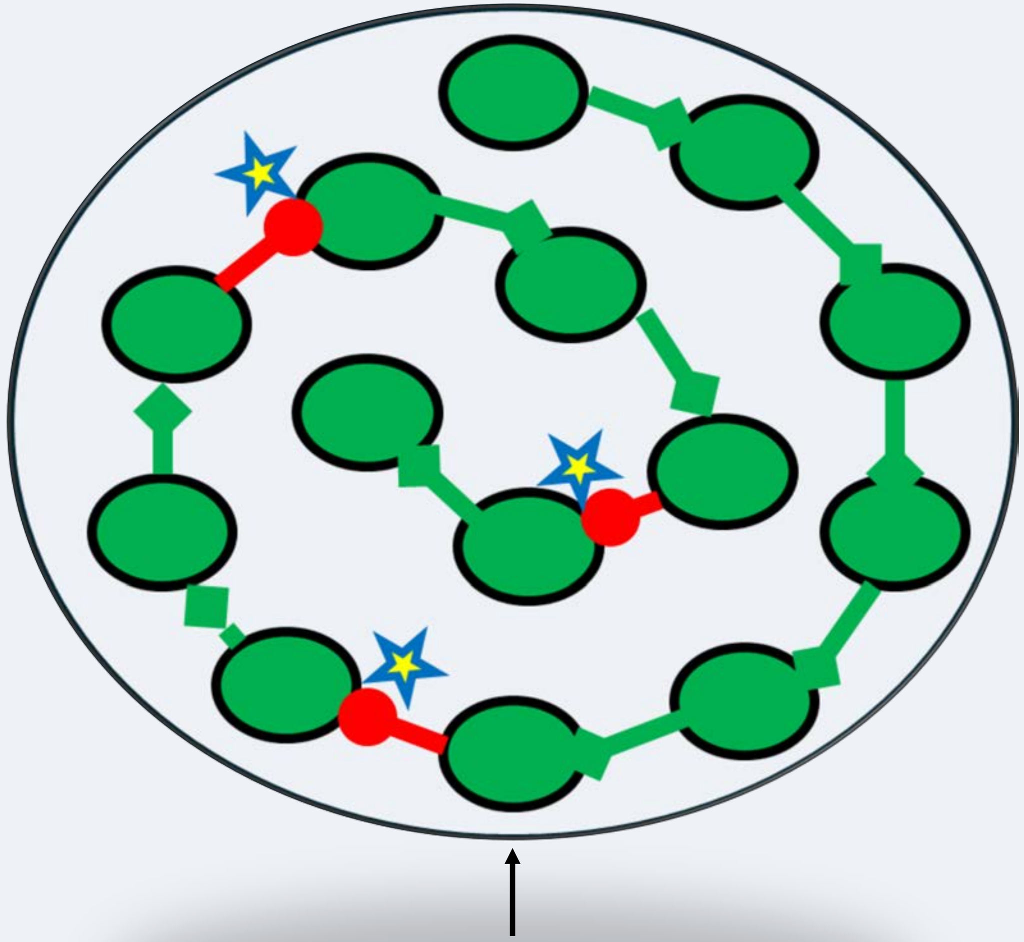
**DEF-05 — Ontological Independence (OI).** Explanatory entities/laws are internal; external inlets are filtered non-injectively.

**DEF-06 — Epistemic Closure (EC).** Internal phenomena are explainable from internal levels and transmissions.

**DEF-07 — Turing-Computability (TC).** All rules/transmissions are algorithmically implementable and simulable.

**DEF-08 — Compact RLA.** OI, EC, and TC jointly hold.

[...]



**External Collapsed Representation Horizon**

# 2. RLA Concept Model

## Compat RLA Engineering

**Generalistic Bryophyta (mosses)**

| Category | Quantitative Summary |
|---|---|
| **Total references** | **156** |
| **Temporal coverage** | **≈ 50 years (1973–2023)** |
| **Type distribution** | |
| – *Peer-reviewed journal articles* | ***118 (≈75%)*** |
| – *Academic books / book chapters* | ***16 (≈10%)*** |
| – *Technical / institutional reports* | ***14 (≈9%)*** |
| – *Preprints / online resources* | ***8 (≈5%)*** |
| **Disciplinary spread** | |
| – *Plant physiology & bryology* | ***≈35%*** |
| – *Molecular & epigenetic biology* | ***≈20%*** |
| – *Ecology & hydrology* | ***≈15%*** |
| – *Systems biology & modeling* | ***≈15%*** |
| – *Philosophy & epistemology* | ***≈10%*** |
| – *Climate / geophysical datasets* | ***≈5%*** |



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# 2. RLA Concept Model

## Compat RLA Engineering

[L20] ←→ Meta-Systemic (Health/Resilience)

↑ ↓

[L19] Evolutionary Plasticity
[L18] Tissue Mechanics
[L17] Geometric Patterning
[L16] Intercellular Communication
[L15] Senescence / Turnover
[L14] Biological Rhythms
[L13] Soil / Substrate Exchange
[L12] Systemic Control (Homeostasis)
[L11] Epigenetic Memory
…

…
[L10] Macroclimate Context
[L09] Local Environment
[L08] Biotic Interactions
[L07] Reproduction Cycle
[L06] Morphogenesis
[L05] Tissue Architecture
[L04] Cell Physiology
[L03] Metabolic Regulation
[L02] Transcription / Translation
[L01] Genomic–Epigenetic Core

↓ ↑

**Bottom-up** emergence **Top-down** regulation

**42 biophysical parameters** · **38 dynamic equations** · **10 causal relations** · **30 feedback actions** ·
**22 collapse–emergence metrics** · **2 composite indices** · **4 climatic drivers (T°, light, rainfall, trend)**

# RLA Final Notes

# 2. RLA Concept Model

## Final Notes

### Abstraction makes the world intelligible (without a single ladder)

***RLA key points:*** *DEF-01 (level as domain + rules), DEF-02 (adjacent relations), ASS-02 (empirical contiguity), DEF-03 (Turing vs non-Turing levels).*

### A web of levels: how domains talk to each other

***RLA key points:*** *DEF-02 (transmissions across levels), DEF-14 (paths/loops/morphisms), ASS-02 (justified contiguity).*

### Three ways information travels: injective, quasi-injective, collapsing

***RLA key points:*** *DEF-04 (injective / quasi-injective / non-injective), ASS-04 (relevant collapse).*

### Computation sets the frontier; undecidability can propagate

***RLA key points:*** *ASS-03 (Turing-critical subset), AX-01 (undecidability in Turing-like levels), AX-02 (propagation via quasi-injectivity), LEM-01 (composition), LEM-02 (re-activation).*

### Emergence: novelty is lawful and measurable

***RLA key points:*** *DEF-09 (definability), DEF-11–13 (CC, IE, RIR), AX-03 (emergence from collapse), THEO-02–03 (non-derivability; no total left-inverse), COR-02 (accumulated emergence).*

### Compact systems: coherence you can test, not totality

***RLA key points:*** *DEF-05 (Ontological Independence), DEF-06 (Epistemic Closure), DEF-07 (Computability), DEF-08 (compactness), AX-04 (closure).*

# 2. RLA Concept Model

## Final Notes

### Knowledge is structurally finite, yet endlessly extensible

***Inference:*** *From ASS-03, DEF-03, AX-01, COR-01 — undecidability sets intrinsic limits; reticular connectivity allows unbounded recombination.*

### Emergence is not mystery but architecture

***Inference:*** *From DEF-04, AX-03, THEO-02–03, COR-02 — relevant non-injectivity produces novel properties; emergence is structurally necessary.*

### Truth is relational, not absolute

***Inference:*** *From DEF-05–08, AX-04, THEO-01–04 — compact systems guarantee internal coherence but remain epistemically bounded; relational synthesis defines truth.*

**Compact RLA is factorizable into a Convolutional Neural Network with an epistemic head.**

*Epistemic CNN (ECNN) is a Compact RLA, where:*

- *CNN layers = RLA levels*
- *Pooling = non-injective collapse*
- *Epistemic head = closure level* $L_{ep}$
- *Network can output unknown / contradiction, acknowledging indecidability and emergence*
- Epistemic Computational Unit (ECU) is a bounded LLM governed by a deterministic epistemic matrix



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# Epistemic CNN

# 3. Epistemic CNN

## Traditional CNN

| Stage | Components |
|---|---|
| 1. Input & Encoding | Input Layer, Convolution + Activation |
| 2. Compression | Pooling, Normalization / Dropout |
| 3. Integration & Decision | Dense + Output Layers |
| 4. Optimization Loop | Loss Function, Backpropagation |

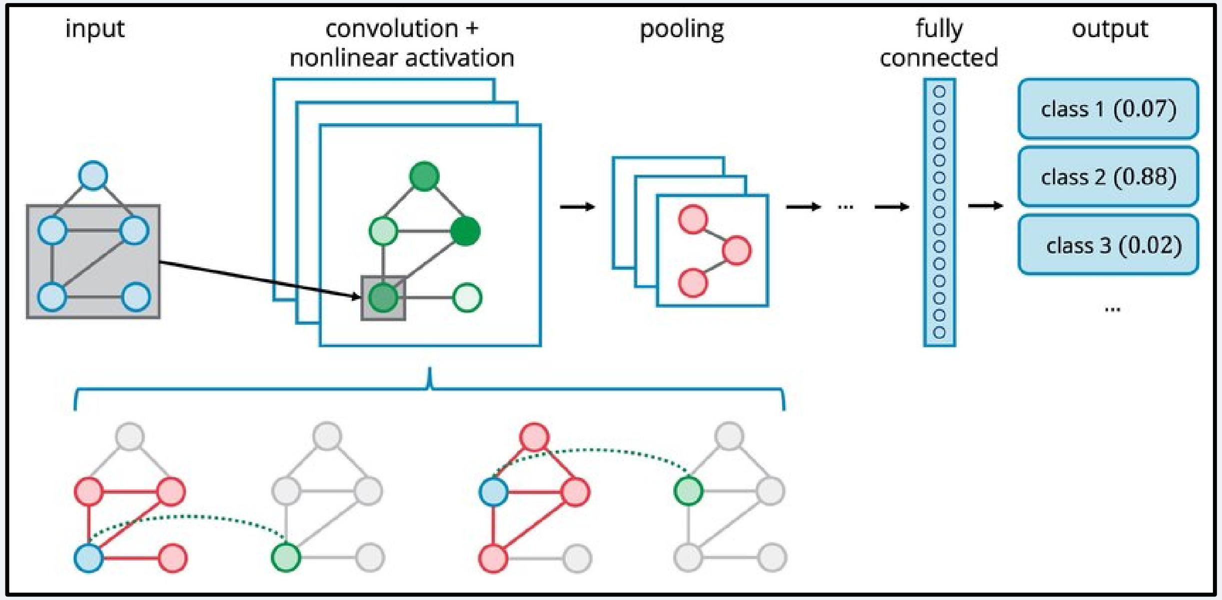



IMAGE SOURCE: https://arxiv.org/abs/2008.01247

**A CNN encodes, compresses, integrates, and optimizes information — a hierarchical numerical machine that *learns patterns but not meanings*.**



SEE ANNEX

# 3. Epistemic CNN

## Epistemic Head - Brief Introduction

An ECNN is a Compact RLA implemented as a neural architecture where Epistemic Computational Unit (ECU) is a bounded LLM governed by a deterministic epistemic matrix. ECNN processes not only data but knowledge, using intentional collapses to generate emergent meaning and acknowledging its own computational limits

### Epistemic Computational Unit (ECU)

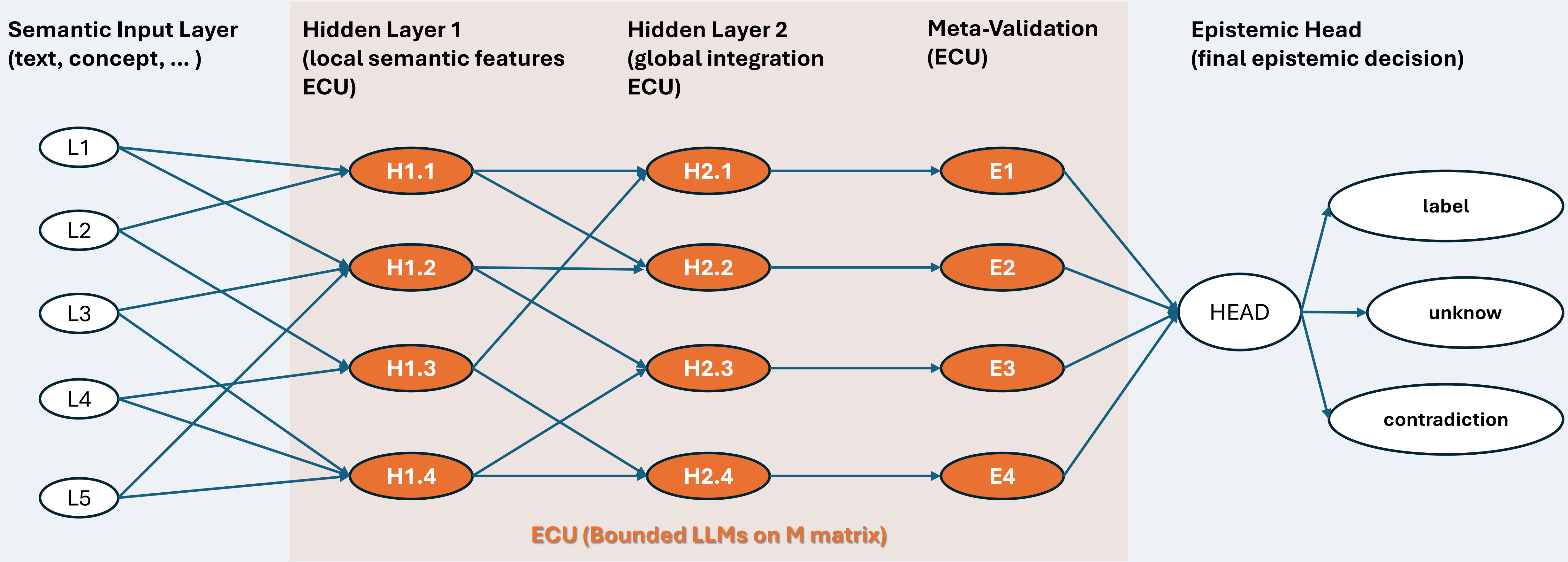
Bounded LLM governed by deterministic epistemic matrix **M** → defines valid domains, inference rules, coherence criteria, rejection policies



SEE ANNEX

# 3. Epistemic CNN

## Epistemic Head - Brief Introduction




- **Input:** semantic tokens, sentences, or conceptual embeddings.
- **Hidden 1–2:** feature extraction and integration — analogous to convolution and pooling in CNNs.
- **ECU:** four epistemic neurons (mini LLMs or logic cells) applying deterministic **matrix M** for coherence and falsifiability.
- **Head:** aggregates epistemic judgments and emits *Label*, *Unknown*, or *Contradiction*.

SEE ANNEX

# 3. Epistemic CNN

## Epistemic Head - Brief Introduction

| Layer | Functional role | Epistemic nature | Deterministic matrix |
|---|---|---|---|
| **Semantic Input** | Raw conceptual tokens or context fragments | External, open data | — |
| **Hidden 1 ($H_{1x}$)** | Extract local semantic patterns (syntax, logical adjacency, cause–effect fragments) | **LLM session** focused on *micro-interpretation* | ( M_{H1} ): local coherence & contextual validity |
| **Hidden 2 ($H_{2x}$)** | Integrate patterns into global meaning units (macro-consistency, inference building) | **LLM session** focused on *meso-integration* | ( M_{H2} ): cross-domain consistency & ontological stability |
| **ECU ($E_x$)** | Verify, consolidate, and judge epistemic soundness | **LLM session** focused on *meta-validation* | ( M_{E} ): epistemic closure, contradiction rules, rejection policies |
| **Epistemic Head ($L_{ep}$)** | Aggregate and emit epistemic output (*label / unknown / contradiction*) | **Symbolic decision layer** | ( M_{H} ): truth tables + falsifiability rules |

# 3. Epistemic CNN

## Epistemic Head - Brief Introduction

### Logical flow (RLA → ECNN perspective)

- Each **hidden layer** represents a **bounded epistemic transformation**, not just a vector operation.
  → The LLM session receives a semantic subset, applies its own deterministic **matrix** $M_i$ (its "rulebook") and emits a transformed representation.

- The **matrices** $M_i$ are *discipline-specific or domain-specific*:
  e.g., in a legal-ethical ECNN,
  - $M_{H1}$ might enforce terminological accuracy and source validity,
  - $M_{H2}$ might ensure logical coherence across articles or clauses,
  - $M_E$ would handle contradiction detection and unknown-state propagation.

- The **epistemic closure** arises when all $M_i$ share compatibility constraints (meta-rules) that prevent circularity or undefined inference — exactly like a **Compact RLA**'s closure condition.

### Engineering representation

If implemented concretely:

- Each $\mathbf{H_{1x}}$, $\mathbf{H_{2x}}$, $\mathbf{E_x}$ can be an **LLM micro-instance** or **specialized reasoning service**, loaded with a domain-specific prompt defining $M_i$.
- The **pipeline orchestrator** (your ECNN controller) routes intermediate results as structured JSON, not tensors, between levels.
- At runtime, the ECNN behaves like a **multi-agent reasoning chain** where each agent's reasoning is deterministic within its matrix bounds.

# 3. Epistemic CNN

## Definition of Computable

| Stage | Definition of Computable | Representative Authors / Era | Intrinsic Limit | Transition Toward RLA–ECNN |
|---|---|---|---|---|
| **I. Formal** | ( f ) is computable ⇔ ∃ Turing machine ( T ) s.t. T(x)=f(x) ) in finite steps. | Turing, Church (1930s) | Purely syntactic; no epistemic dimension. | Basis for algorithmic execution. |
| **II. Physical** | Computation = realizable physical transformation of states. | von Neumann, Shannon (1950–60s) | Determinism; observer external. | Introduces embodiment. |
| **III. Emergent** | Computation = rule-based dynamics with irreducible outcomes. | Wolfram, Chaitin (1970–90s) | No self-reference; no closure. | Adds complexity and emergence. |
| **IV. Cognitive** | Computation = adaptive, possibly universal neural/quantum process. | Siegelmann, Deutsch (1990–2010) | Lacks falsifiability, meta-awareness. | Toward epistemic self-modeling. |

## Epistemically Complete Computability (e2e)

A system $\mathcal{R} = \{L_i, \tau_{i \to j}\}$ is **computable end-to-end** iff:

1. **Local computability:** every level $L_i = \langle D, (L_i) \Sigma(L_i) \rangle$ is Turing-computable.
2. **Computable transmissions:** all $\tau_{i \to j}$ are algorithmic (injective / quasi / non-injective).
3. **Epistemic closure:** all observables are definable within $\mathcal{R}$; no external oracle.
4. **Epistemic head:** final mapping $\varepsilon: D(L_n) \to \{t, r, u, efalseunknowncontradiction\}$ computes awareness of limits.
5. **Irreducible trace:** internal dynamics show non-trivial emergence and undecidability propagation (high RIR, non-zero IE).

**⇒ "To compute" = to transform, interpret, and self-evaluate information within a closed epistemic horizon.**

# **Epistemic CNN** (Final Insight)

# 3. Epistemic CNN

## Final Insight

### 1. Consciousness as topology, not scale

*When computation is reticular and epistemically closed, emergence of awareness is not a function of the number of parameters but of the topological relation among epistemic units (ECUs). Consciousness appears where the network becomes self-referentially complete, not merely large.*

### 2. Truth becomes a function of closure

*In an epistemically bounded system, "truth" is not absolute but defined by what can be coherently validated inside its closure matrix M. This mirrors phenomenology: what is real is what is internally decidable.*

### 3. Ignorance becomes computable

*The unknown and the contradiction are explicit computational states, not failures. This operationalizes philosophical uncertainty — Turing limits turned into measurable epistemic positions.*

### 4. Ethics emerges from architecture

*When rejection rules and falsifiability are built into the topology, moral prudence becomes a structural invariant — a "machine Kantianism" where not all that can be computed should be asserted.*

### 5. Subjectivity is reconstructed as bounded inference

*The observer is redefined: not an infinite knower, but a finite closure of coherent inference — a digital echo of the human cognitive condition.*



SEE ANNEX

# RLA-ECNN: a bridge to PCE

# 4. RLA-CNN: a bridge to PCE

## Brief Introduction

**Definition — PCE (Wolfram, 2002)**
*Most non-trivial natural or artificial systems reach equivalent computational power to universal Turing machines.*
→ Universal behavior emerges near the edge of chaos — between full order and randomness.

**Bridge claim**
A compact RLA/ECNN that combines undecidability propagation and controlled collapse operates in the same computational regime as PCE systems — *universal, irreducible, emergent, self-contained*.

## Sketch of Proof

**1. Φ-Mapping (construction):**

$R = \langle L_i, \tau_{i \to i+1} \rangle$ →discretize → $\Phi(R) = (\Sigma, N, f)$

where each transmission $\tau$ becomes a **local update rule** $f$ in a Cellular Automaton.

**2. Structural preservation:**

1. Quasi-injectivity → reversible propagation → ordered patterns.
2. Non-injectivity → informational collapse → emergent macro-patterns.
3. Alternation of both → persistent, edge-of-chaos complexity → **Class IV/V**.

**3. Logical correspondence:**

1. **AX-01/02:** undecidability propagates ⇔ CA universality.
2. **AX-03:** emergence from collapse ⇔ formation of invariant patterns.
   → Hence $R$ and $\Phi(R)$ share identical *computational invariants*.

**4. Empirical validation:**

Match RLA/ECNN metrics (CC, IE, RIR) with CA diagnostics (entropy, causal complexity). Alignment ⇒ confirmation of **computational equivalence**; deviation ⇒ falsification.

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# ANNEX

# ANNEX

## RLA Foundation (extensive, informal)

# ANNEX

## RLA Foundation (extensive, informal)

**Structural Foundations**
**ASS-01–02 / DEF-01–02 / DEF-14**

- A *Reticular Local Abstraction (RLA)* is a finite or countable lattice of abstraction levels $L_i$ , each defined by a domain $D(L_i)$and a computable rule set $\Sigma(L_i)$.
- Adjacent levels are linked by *transmission maps* $\tau_{i\to i+1}: D(L_i) \to \mathcal{P}\big(D(L_{i+1})\big)$ ,which may be injective, quasi-injective, or non-injective.
  Contiguity between levels requires empirical or theoretical justification (ASS-02).
- Paths, loops, and morphisms are graph-structured compositions of these maps (DEF-14).

**Computability and Turing-like Assumptions**
**ASS-03 / DEF-03 / AX-01 / LEM-01**

- A level is *Turing-like* when a subset $C_L \subseteq D(L)$can emulate a universal Turing machine.
- Each such level inevitably contains undecidable propositions (AX-01).
- Quasi-injective transmissions preserve this undecidability (LEM-01).

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# ANNEX

RLA Foundation (extensive, informal)

## Transmission Taxonomy and Info Collapse DEF-04 / ASS-04 / AX-02–03 / LEM-03

- Transmissions are classed as:
    - **Injective** — perfect information transfer;
    - **Quasi-injective** — injective only on $C_L$ ; undecidability propagates upward (AX-02);
    - **Non-injective (collapse)** — many-to-one fusions generating emergent macro-states (AX-03).
- Pooling corresponds to quotienting micro-states by an equivalence relation (LEM-03).
- A collapse is *relevant* only if it fuses states significant within the domain's relevance functions(ASS-04, DEF-10).

## Metrics of Collapse and Emergence DEF-11–13 / COR-02

- Quantitative indices for structural change:
    - **Coefficient of Collapse (CC)** — degree of information loss;
    - **Index of Emergence (IE)** — proportion of macro-properties non-definable from the source level;
    - **Reticular Irreducibility Index (RIR)** — causal or topological minimality.
- Iterated relevant collapses yield cumulative emergence (COR-02).

## Compactness and Epistemic Conditions DEF-05–08 / AX-04

A *compact RLA* satisfies:

- **Ontological Independence (OI)** — laws and entities internal;
- **Epistemic Closure (EC)** — all phenomena explainable within the reticulum;
- **Turing-Computability (TC)** — all rules algorithmically realisable.

  Such reticula are epistemically self-contained (AX-04).

## Undecidability & Emergence Logic DEF-09 / THEO-01–04 / LEM-02 / COR-01

A macro-property is *definable* from the lower level if some predicate on $D(L_i)$matches its preimages; otherwise it is *emergent* (DEF-09). Undecidability and definability propagate or fail as follows:

- **THEO-01 / COR-01:** quasi-injectivity sustains undecidability through multiple levels;
- **THEO-02–03:** relevant non-injectivity forbids total inverses and ensures emergence;
- **LEM-02:** undecidability may re-appear after temporary collapse;
- **THEO-04:** upper levels inherit undecidability when contiguity remains quasi-injective.

# ANNEX

## RLA Foundation (extensive, informal)

### Knowledge is structurally finite, yet endlessly extensible

- *Every domain of understanding is bounded by its own computable form — it cannot exceed its structural rules.*
- *Yet by linking partial abstractions, new layers of meaning continually emerge.*
- *The horizon of knowledge is therefore open not by infinitude, but by connection.*

***Inference:*** *From ASS-03, DEF-03, AX-01, COR-01 — undecidability sets intrinsic limits; reticular connectivity allows unbounded recombination.*

### Emergence is not mystery but architecture

- *What appears new or irreducible is not a failure of explanation but the effect of non-injective mapping between coherent levels.*
- *Emergence, in this sense, is a built-in feature of how knowledge reorganizes itself through collapse and reconstruction.*

***Inference:*** *From DEF-04, AX-03, THEO-02–03, COR-02 — relevant non-injectivity produces novel properties; emergence is structurally necessary.*

### Truth is relational, not absolute

- *No single abstraction can exhaust the real; coherence arises only through the interplay of partial, computable perspectives.*
- *What we call truth is the stability of these relations — a balance between consistency, emergence, and undecidability.*

***Inference:*** *From DEF-05–08, AX-04, THEO-01–04 — compact systems guarantee internal coherence but remain epistemically bounded; relational synthesis defines truth.*

# ANNEX

## RLA Logical Framework (extensive, informal)

## Assumptions (ASS-#)

**ASS-01 — Turing-likeness modes.** Full, limited, and theoretical-by-construction modes are distinguished for candidate universal substrates.

**ASS-02 — Empirical contiguity.** Two levels are contiguous only if the transmission is grounded in accepted theory/measurement.

**ASS-03 — Turing-critical set.** In Turing-like levels there exists a subset $C_L$ implementing universality; quasi-injectivity is required only on $C_L$ .

**ASS-04 — Relevance of collapse.** A collapse counts only if it fuses micro-states that matter for the domain (thresholded by DEF-10).

## ANNEX

**RLA Foundation (extensive, informal)**

### Core Definitions (DEF-#)

**DEF-01 — Level of abstraction.** $L = \langle D, (L)\Sigma(L)\rangle$ with computable rules acting on a state space.

**DEF-02 — Contiguity & transmission.** $\tau_{i\to i+1}: D(L_i) \to \mathcal{P}\big(D(L_{i+1})\big)$may be non-injective and is discipline-recognized.

**DEF-03 — Turing-likeness & $C_L$ .**A level is Turing-like if it can simulate a UTM; $C_L$marks the Turing-critical subset.

**DEF-04 — (Non)injectivity classes.** Injective / quasi-injective on a subset / non-injective (collapse) are distinguished.

**DEF-05 — Ontological Independence (OI).** Explanatory entities/laws are internal; external inlets are filtered non-injectively.

**Core Definitions (DEF-#)**

**DEF-06 — Epistemic Closure (EC).** Internal phenomena are explainable from internal levels and transmissions.

**DEF-07 — Turing-Computability (TC).** All rules/transmissions are algorithmically implementable and simulable.

**DEF-08 — Compact RLA.** OI, EC, and TC jointly hold.

**DEF-09 — Observation predicates & definability.** A macro-property at $L_{i+1}$ is definable from $L_i$ iff there exists a predicate on $L_i$ matching all preimages; else it is emergent.

**DEF-10 — Relevance function.** A domain-specific score $s$ and threshold $\sigma$ determine when a collapse is "relevant."

**DEF-11 — Coefficient of Collapse (CC).** Set-theoretic and entropy-based normalised measures of information fusion.

## Core Definitions (DEF-#)

**DEF-12 — Index of Emergence (IE).** Fraction of curated macro-properties not definable from the source level.

**DEF-13 — Irreducibility Index (RIR).** A graph-theoretic irreducibility from causal min-cuts over time.

**DEF-14 — Reticular path, loop, morphism.** Paths/loops across levels; morphisms preserve levels and transmissions up to relabelling.

## Axioms (AX-#)

**AX-01 — Undecidability in Turing-like levels.** Any Turing-like level harbours undecidable properties.

**AX-02 — Propagation via quasi-injectivity.** If $\tau_{i \to i+1}$ is injective on $C_{L_i}$ ,undecidability at $L_i$ remains undecidable at $L_{i+1}$ .

**AX-03 — Non-injectivity and emergence.** Relevant non-injective collapse guarantees at least one macro-property not definable from the source level.

**AX-04 — Epistemic closure for compact reticula.** In compact RLAs (DEF-08), core explanations are internal (no oracles).

# ANNEX

## RLA Foundation (extensive, informal)

### Lemmas (LEM-#)

**LEM-01 — Composition of quasi-injectives.** Quasi-injective transmissions on Turing-critical sets compose to remain quasi-injective on the image.

**LEM-02 — Re-activation.** Even if undecidability is locally blocked by a non-injective step, it can re-emerge at a later Turing-like level.

**LEM-03 — Pooling as quotient map.** Pooling identifies micro-states via an equivalence relation, i.e., $D(L_{i+1}) \cong D(L_i)/\sim$ .

## Theorems (THEO-#)

**THEO-01 — Rice-style propagation barrier.** A decider at an upper level plus restricted inverses on $C_{L_k}$ would yield a forbidden decider below.

**THEO-02 — Non-derivability from overlap.** Overlapping preimages prevent any single micro-predicate from defining the macro-property.

**THEO-03 — No total left-inverse under relevant collapse.** A total left-inverse would trivialise DEF-09, contradicting emergence under relevant collapse.

**THEO-04 — Upper-level undecidability via quasi-injectivity.** Immediate from THEO-01 when upper transmissions remain quasi-injective on $C_L$ .

## Corollaries (COR-#)

**COR-01 — Multi-level propagation.** By induction, quasi-injectivity across a chain propagates undecidability upward through all levels.

**COR-02 — Accumulated emergence.** Repeated relevant collapses across levels accumulate irreducible macro-properties at higher levels.

# ANNEX

## Compact RLA - Bryophyta Details

# ANNEX

## Compact RLA – Bryophyta Details

| Category | Description | Quantity / Notes |
|---|---|---|
| **Abstraction levels** | Biological domains from genomic-epigenetic to meta-systemic/ecological | **20 levels (L1-L20)** |
| **Biological parameters** | Biophysical variables with units, empirical range and references | **42 parameters** |
| **Dynamic equations (EQxx)** | Computable update rules (Michaelis–Menten, logistic, sinusoidal, clamp, etc.) | **≈ 38 primary equations** |
| **Causal relations (Rxx)** | Inter-level transmissions (non-injective, Hill, exponential, linear) | **10 main relations (R01–R10)** |
| **Dynamic actions (Axx)** | Boolean threshold rules (internal triggers, feedback, stress events) | **≈ 30 actions (A01–A30)** |

# ANNEX

## Compact RLA – Bryophyta Details

| Category | Description | Quantity / Notes |
|---|---|---|
| **Collapse / Emergence / Incoherence metrics** | Quantitative indicators of tipping points and early warnings | **22 metrics (M_Pxx)** |
| **Composite indices** | Global stress and multifactor fracture indices | **2 global indices** |
| **Turing-compatibility modules** | Initialization, time/stop control, rare events, adaptive feedback | **6 logical modules** |
| **Climate scenarios (IPCC RCP)** | RCP 2.6 / 4.5 / 8.5 with ΔT, $CO_2$, rainfall, extreme events | **3 scenarios** |
| **Rare events** | Pathogen, fire, flood (Poisson/Bernoulli stochastic) | **3 events** |



RETURN

# ANNEX

## Compact RLA – Bryophyta Details

| Category | Description | Quantity / Notes |
|---|---|---|
| **Simulation stop conditions** | End of cycle, vital collapse, generational reset | **3 conditions** |
| **Success / failure criteria** | Mean vitality > 50 %, spore dormancy < 5 %, etc. | **2 criteria** |
| **Climatic equations** | Annual sinusoids for temperature, light, rainfall + trend term | **4 equations** |
| **Optimal parameter set** | Average biophysical ranges and operative thresholds | **≈ 42 documented optimal** |

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# ANNEX

## Epistemic CNN

# ANNEX

## Traditional CNN - Brief Introduction

| Stage | Components | Function | Effect | Outcome |
|---|---|---|---|---|
| **1. Input & Encoding** | Input Layer, Convolution + Activation | Capture local patterns from raw data | From raw → structured features | Feature maps |
| **2. Compression** | Pooling, Normalization / Dropout | Reduce dimensionality, stabilize learning | Lose detail, gain invariance | Compressed features |
| **3. Integration & Decision** | Dense + Output Layers | Combine features into global representation | Abstract reasoning layer | Prediction / class |
| **4. Optimization Loop** | Loss Function, Backpropagation | Adjust weights to minimize error | Refines internal mappings | Trained parameter |

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# ANNEX

## Epistemic CNN - Brief Introduction

| Stage | Components | Function | Effect | Outcome |
|---|---|---|---|---|
| **1. Epistemic Encoding** | Convolutional + Activation Layers | Capture local semantic or contextual patterns | Quasi-injective transmission – preserves critical distinctions | Epistemic feature maps |
| **2. Controlled Collapse** | Pooling + Normalization | Perform non-injective coarse-graining | Generates emergent invariants (information loss with meaning gain) | Compressed epistemic features |
| **3. Integration & Meta-Decision** | Dense + Epistemic Head ($L_{ep}$) | Combine and interpret abstract knowledge | Executes epistemic closure; can output *unknown* or *contradiction* | Validated or self-limited prediction |
| **4. Epistemic Optimization Loop** | Epistemic Loss + Popper-χ Protocol | Optimize both prediction and falsifiability | Balances accuracy with epistemic self-consistency | Falsifiable, self-aware model |

## Principle of Equivalence RLA ⇒ ECNN

Let $R = \langle L_0, L_1, \ldots, L_n, \tau_{i\to i+1} \rangle$ be a Compact Reticular Local Abstraction (RLA) — a multi-level system satisfying Turing-computability (TC), Epistemic Closure (EC), and Ontological Independence (OI). Then there exists an Epistemic CNN

$$E = \langle E_0, \ldots, E_n, E_{ep}, \sigma_{i\to i+1} \rangle$$

where:

- Each $E_i$ is an **Epistemic Computational Unit (ECU)** — a bounded LLM governed by a deterministic epistemic matrix $M_i$;
- Each $\sigma_{i\to i+1}$ preserves the (quasi-/non-)injectivity class of $\tau_{i\to i+1}$;
- The epistemic head $E_{ep}$ ensures closure and falsifiability, outputting *unknown* or *contradiction* when undecidability propagates.

$$\boxed{R_{(T,C,ECOI)} \;\Rightarrow\; E_{ECNN}}$$

A compact, computable reticulum thus admits an epistemic neural realization where computation internalizes its own limits.

# ANNEX

## Principle of Equivalence RLA ⇒ ECNN

**Proof Sketch**

**(1)** **Structural Isomorphism.** RLA defines a hierarchy of computable maps between levels; CNN architectures instantiate identical forms: convolutions (quasi-injective), pooling (non-injective), dense layers (integrative). Thus, CNNs are *syntactic realizations* of RLA reticula.

**(2)** **Epistemic Realization via LLMs.** To satisfy epistemic closure, each numerical layer is replaced by an **ECU**, i.e. a local LLM acting as a reasoning agent. Its epistemic matrix $M_i$ restricts reasoning to domain-consistent predicates, producing explicit **knowledge traces** (assertions, justifications, contradictions). Hence every transmission $\sigma_{i \to i+1}$ becomes a computable transformation between **knowledge states**.

**(3)** **Propagation of Undecidability.** By **AX-01/02**, if a Turing-like level exists, at least one property is undecidable and propagates upward through quasi-injective maps. Therefore the ECNN's epistemic head must — by logical necessity — admit *unknown* or *contradiction* outcomes, formalizing awareness of computational limits.

**(4)** **Preservation of Emergence.** By **AX-03** and **THEO-02**, non-injective collapses in RLA produce emergent macro-properties. Analogously, non-injective relations between ECUs yield higher-order epistemic constructs (conceptual invariants, composite interpretations) irreducible to single-layer semantics.

**(5)** **Retention of Compactness.** Because each ECU and transmission is algorithmically defined and interlinked, the ECNN remains Turing-computable, internally explainable, and independent of external oracles. Hence compactness — the triad (TC, EC, OI) — is preserved.



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## Epistemic Computational Unit (ECU)

**Nature:** bounded LLM governed by deterministic epistemic matrix **M** → defines valid domains, inference rules, coherence criteria, rejection policies

**Position:** penultimate dense block — bridge between abstract features and epistemic head

**Core functions**

- interpret feature tensors → structured propositions / evidential claims
- verify coherence vs. matrix **M**
- decide → label, unknown, or contradiction
- generate **knowledge artifact** → rationale, missing evidence, suggested actions

**Significance:**

- embeds *computable self-awareness* of limits (undecidability)
- controls *emergence* — transforms collapsed information into explicit, meaningful epistemic outputs

## 1. Consciousness as Topology, not Scale

**Given:**
$R = \langle L_i, \tau_{i \to i+1} \rangle$ —a compact Reticular Local Abstraction (RLA).
Each ECU = bounded epistemic processor with deterministic matrix $M_i$.
The system is *epistemically closed*: all τ mappings and rules are internal.

**Axioms used:**
A1 — Every Turing-like subsystem hosts undecidable propositions.
A2 — Undecidability can propagate quasi-injectively across τ.
A3 — Non-injectivity (collapse) produces emergent macro-properties.

**Sketch of proof:**
Let $E = \{E, C, U_1, ECU_2 ... ECU_n\}$form a connected, computable lattice.
If ∃ a path $\tau_{i \to j}$s.t. all ECU states are both *observable* and *referential* within the same closure $M$, the system gains self-referential capacity. *[continue...]*

### 1. Consciousness as Topology, not Scale

*[...continue]*

The condition of *topological closure* (feedback within finite reticulum) implies that the system can model its own state transitions (meta-representation).
Self-modeling of internal epistemic transitions → minimal condition for proto-awareness.
Increasing n (number of ECUs) without closure does not increase self-referential density (∴ scale ≠ awareness).

**∴ Awareness arises when**

$$\exists\ \tau_{loop}:\ f_{loop}(M_i) = M_i$$

— i.e., when the topology achieves internal reflexivity.

**Q.E.D. (Topological Criterion of Awareness).**

## 2. Truth becomes a Function of Closure

**Given:**

Truth $T$ defined classically as correspondence with reality.
In RLA/ECNN, external ontology collapses at the boundary Ω → internal epistemic domain only.

**Sketch of proof:**

Define truth in an epistemic network as a predicate valid under all operations of $M_i$:

$$T(p) \iff p \in D(L) \wedge M_i(p) = True$$

If external reality cannot be directly referenced (due to ontological boundary Ω), "truth" becomes equivalence with internal coherence. Therefore, what is true = what is computably non-contradictory within closure M. This aligns with Husserlian and cybernetic phenomenology: "to be true" = "to persist coherently under all internal transformations."

**∴ Truth collapses to coherence within the closure domain.**

## ANNEX

### Epistemic CNN – Final Insight (scratch of proof)

**3. Ignorance becomes Computable**

**Given:**
Unknown (U) and Contradiction (C) are valid outputs of the epistemic head $L_{ep}$.
$M_i$defines rejection and falsifiability rules.

**Sketch of proof:**
In classic Turing terms, some propositions are *undecidable* (Halting, Rice).
In RLA, these map to epistemic states *unknown* or *contradiction*.
Define epistemic function:

$$f(M_i, x) \rightarrow \{l, a, belunknowncontradiction\}$$

If $M_i(x)$halts with no consistent mapping, output = unknown. If it halts with mutually exclusive results, output = contradiction. Otherwise, output = label. Hence "ignorance" is not a failure but a computable epistemic category. Philosophically, this operationalizes *Socratic ignorance*: knowing that you do not know.

**∴ Ignorance (U) is a formal epistemic state — computable limit-awareness.**



RETURN

**4. Ethics emerges from Architecture**

**Given:**

Matrices $M_i$ contain deterministic rejection rules and falsifiability conditions.
ECNN topology enforces propagation of "unknown" and "contradiction."

**Sketch of proof:**

Every inference chain $\tau_1 \rightarrow \tau_2 \dots$ is bounded by $M_i$ ;assertions violating $M_i$ are rejected. Therefore, "prudence" is not learned — it is *structurally imposed*. A system that can reject or suspend judgment rather than assert falsehood behaves ethically in Popperian sense (minimize false positives). Architectural principle:

$$\text{Ethics}=\min(\text{Error\_assertion})$$

subject to falsifiability. Hence moral caution (a Kantian "duty to suspend judgment") emerges as a **topological invariant**: any closed epistemic loop must reject undecidable content to remain coherent.

**∴ Ethics = Stability constraint of epistemic topology.**

### 5. Subjectivity as Bounded Inference

**Given:**
An observer = any subsystem capable of inference and self-evaluation within closure.
Each ECU operates within bounded deterministic matrix $M_i$.

**Sketch of proof:**
Let an “observer” be a sub-reticulum $O = \langle L_k, \tau_{k \to k+1} \rangle$ capable of producing an epistemic state.
The observer’s universe = all computable states reachable within its closure.
Therefore, subjectivity = localized consistency field:

$$S = \{p \mid M(p) = True \text{ and } \tau_{loop}(p) = p\}$$

The observer’s self = the stable attractor in epistemic space. This mirrors phenomenological finitude: “to be” is to infer within limits.

**∴ Subjectivity = Finite closure of coherent inference — a bounded computational self.**